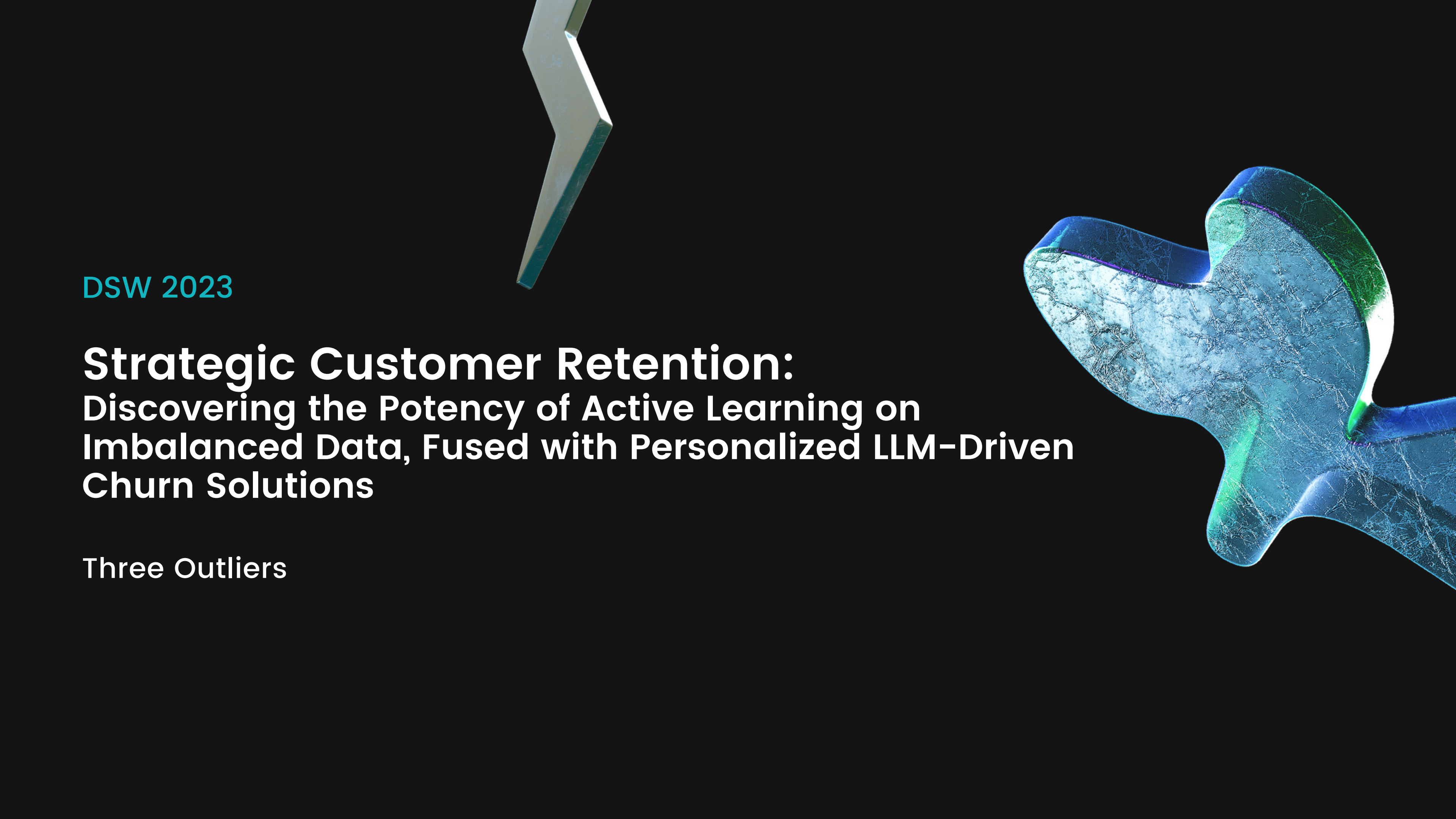




DSW 2023

Strategic Customer Retention: Discovering the Potency of Active Learning on Imbalanced Data, Fused with Personalized LLM-Driven Churn Solutions

Three Outliers

Meet the **Three Outliers**



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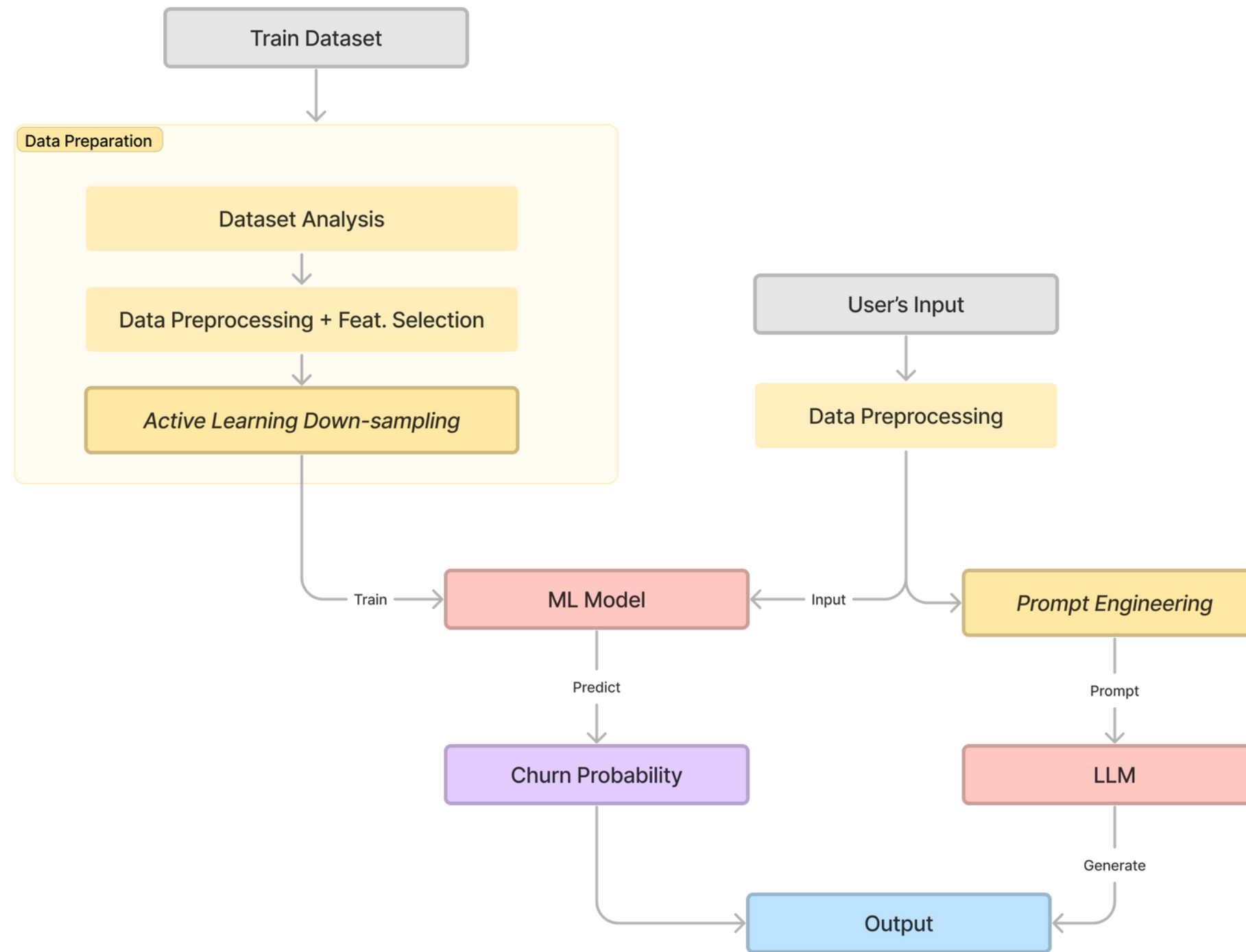
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Background

Telecom company faces the challenge of predicting churn among high-value customers with a dataset of diverse customer attributes.

To tackle this challenge, we implement an innovative model that utilizes active learning, thereby ensuring a balanced evaluation of precision and recall in the performance assessment of our model. Additionally, we proactively address the potential for churn by generating customized solutions that cater to the specific needs of at-risk customers.

Proposed Solution Flow

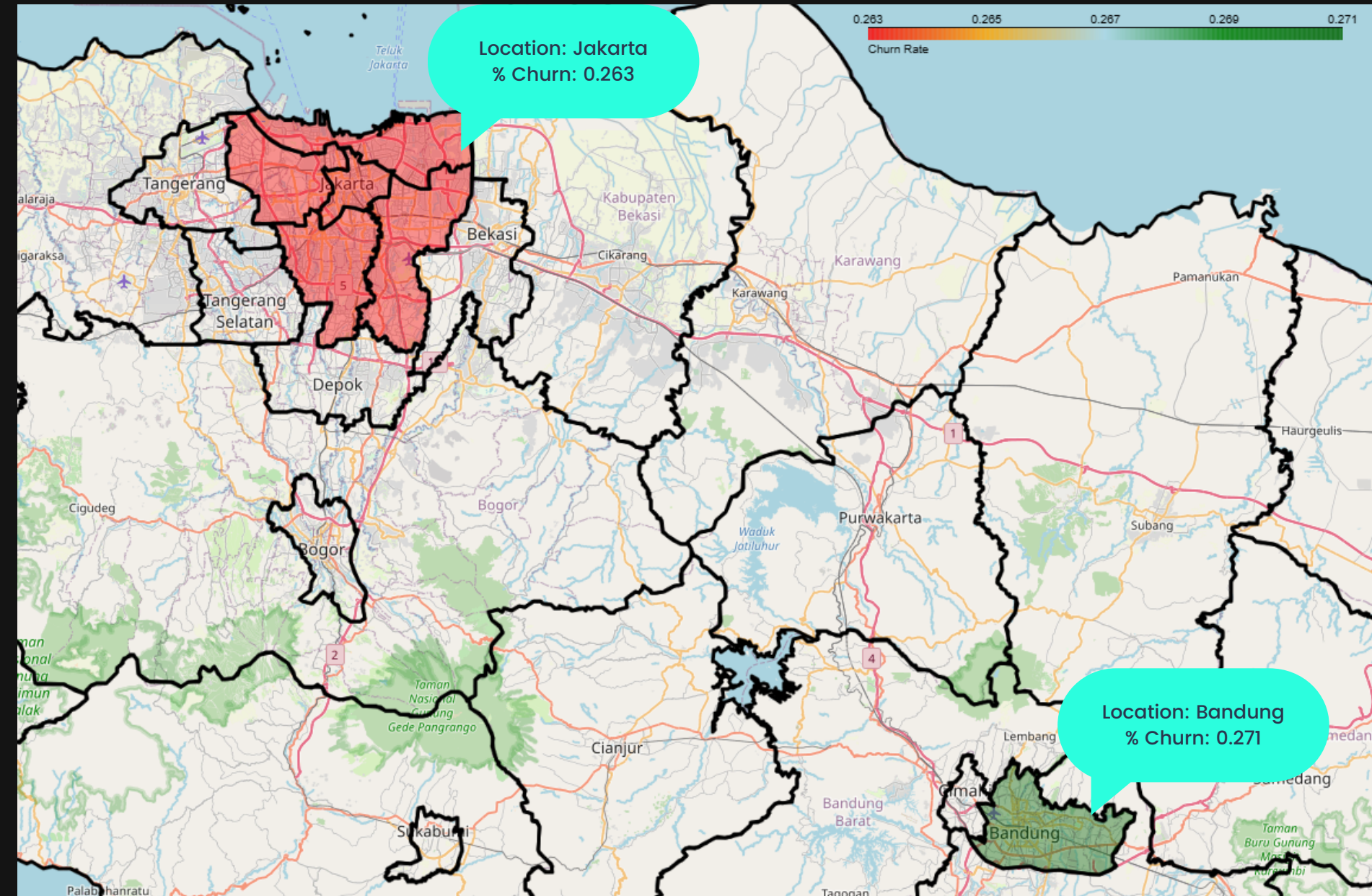
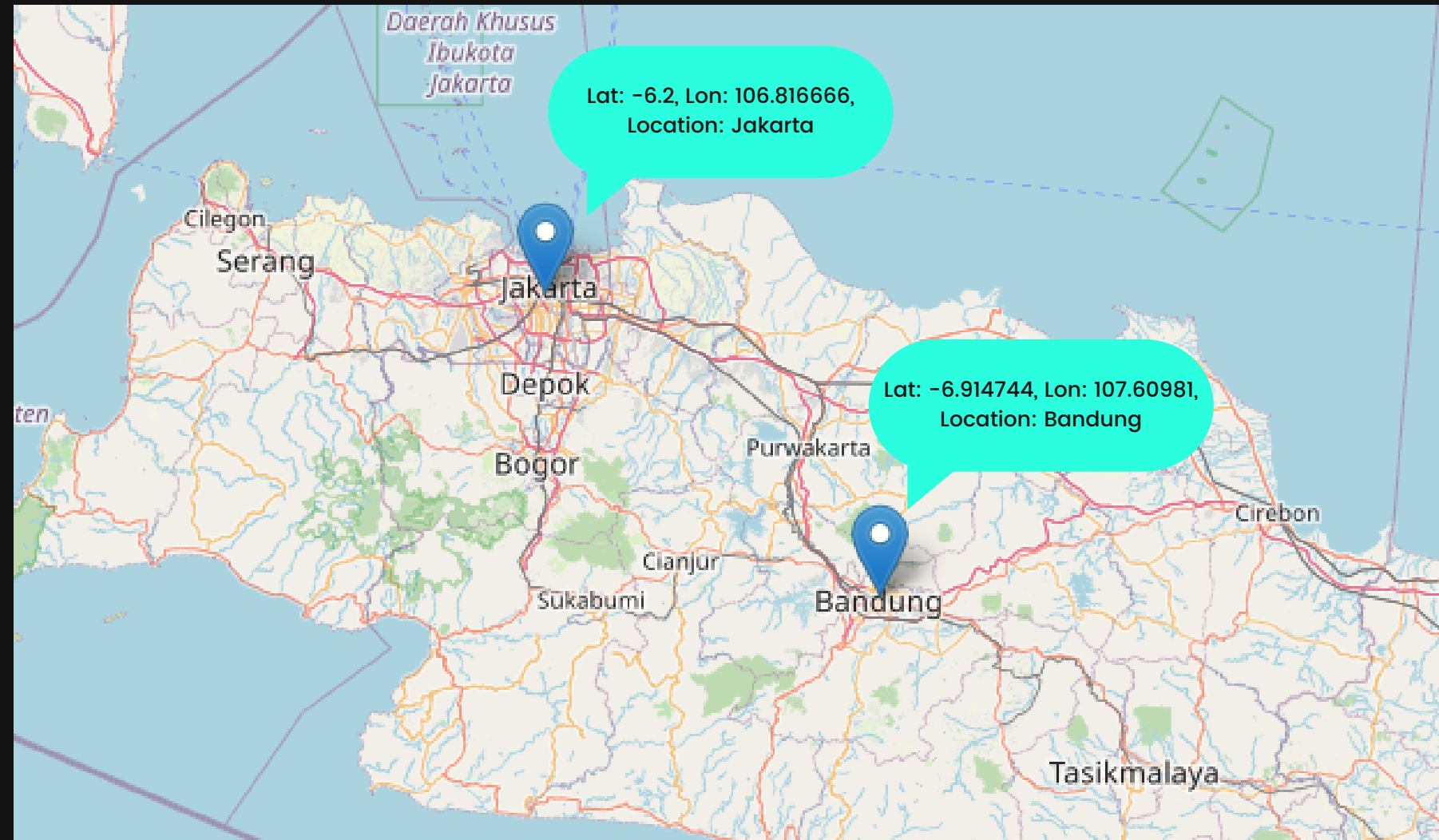


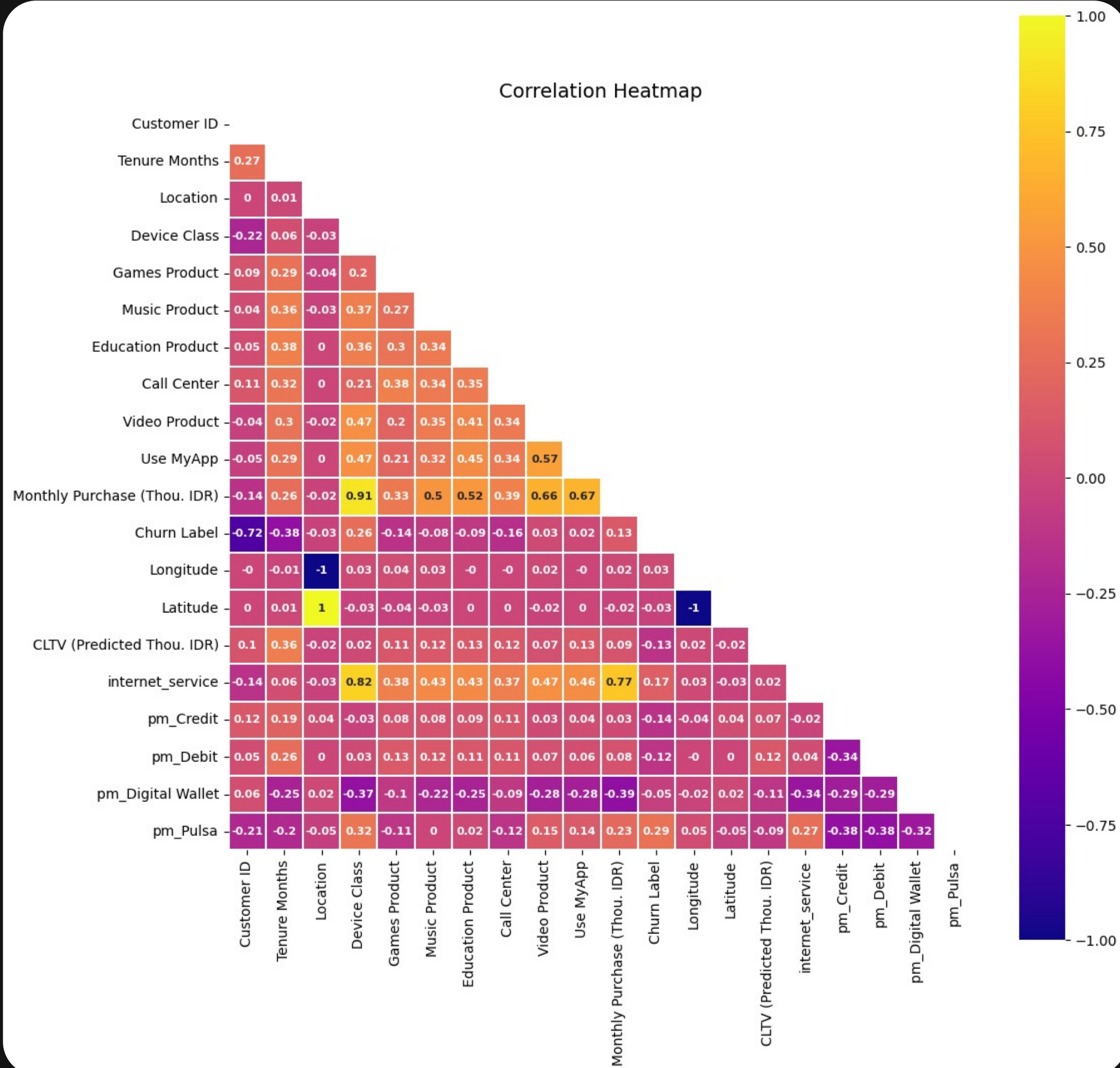
Dataset Overview

- Key Features: Unique customer IDs, tenure, location, device class, and indicators for gaming, music, and education product usage.
- Engagement Metrics: Call center, video services, proprietary MyApp, payment methods, and monthly expenditures.
- Churn Indication: Binary label indicating customer departure in the specified quarter.
- Geospatial Data: Longitude and latitude coordinates.
- Strategic Metric: Customer Lifetime Value (CLTV) in thousands of IDR.

Customer Locations

There are only two locations in the dataset!

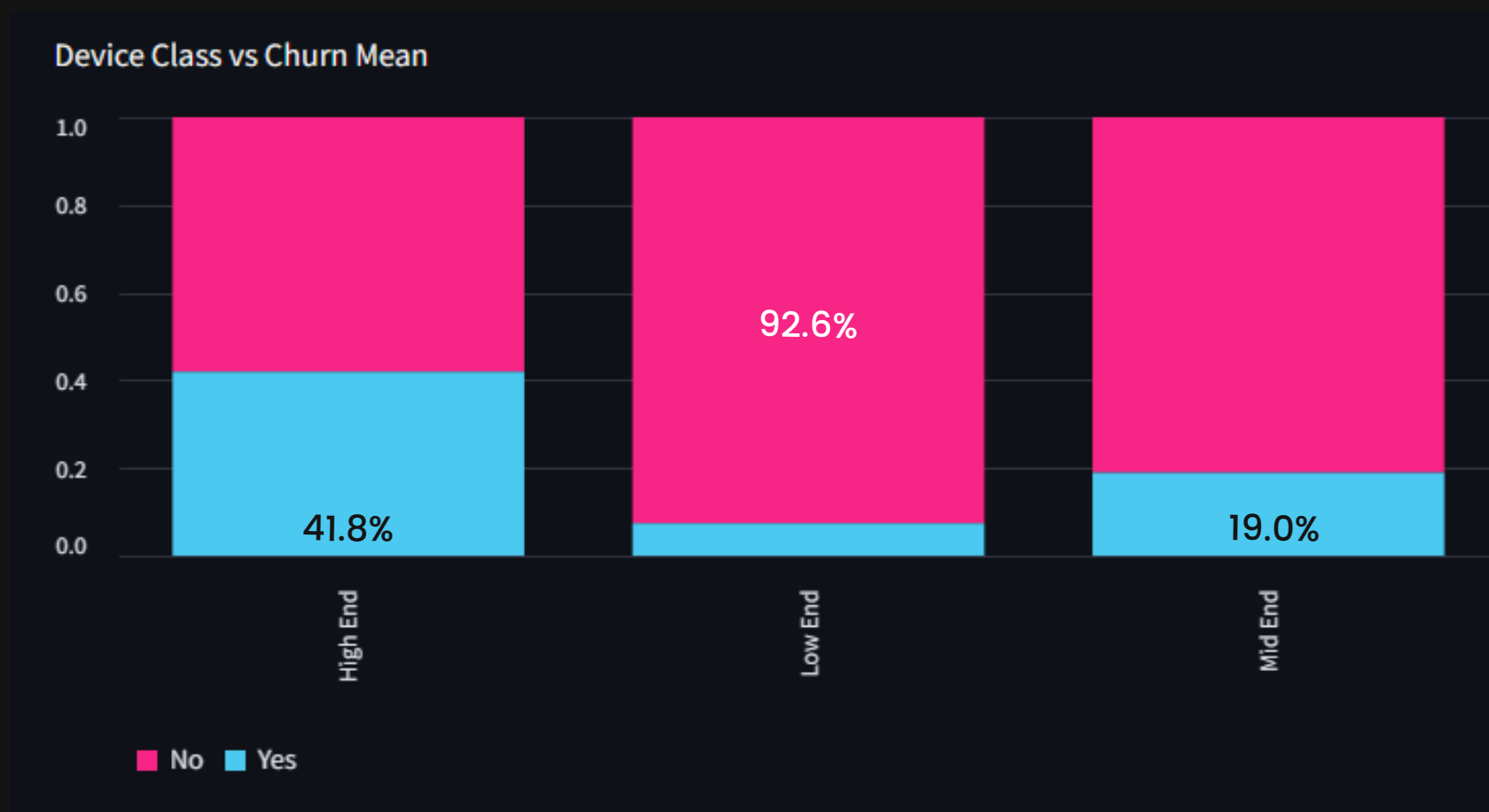
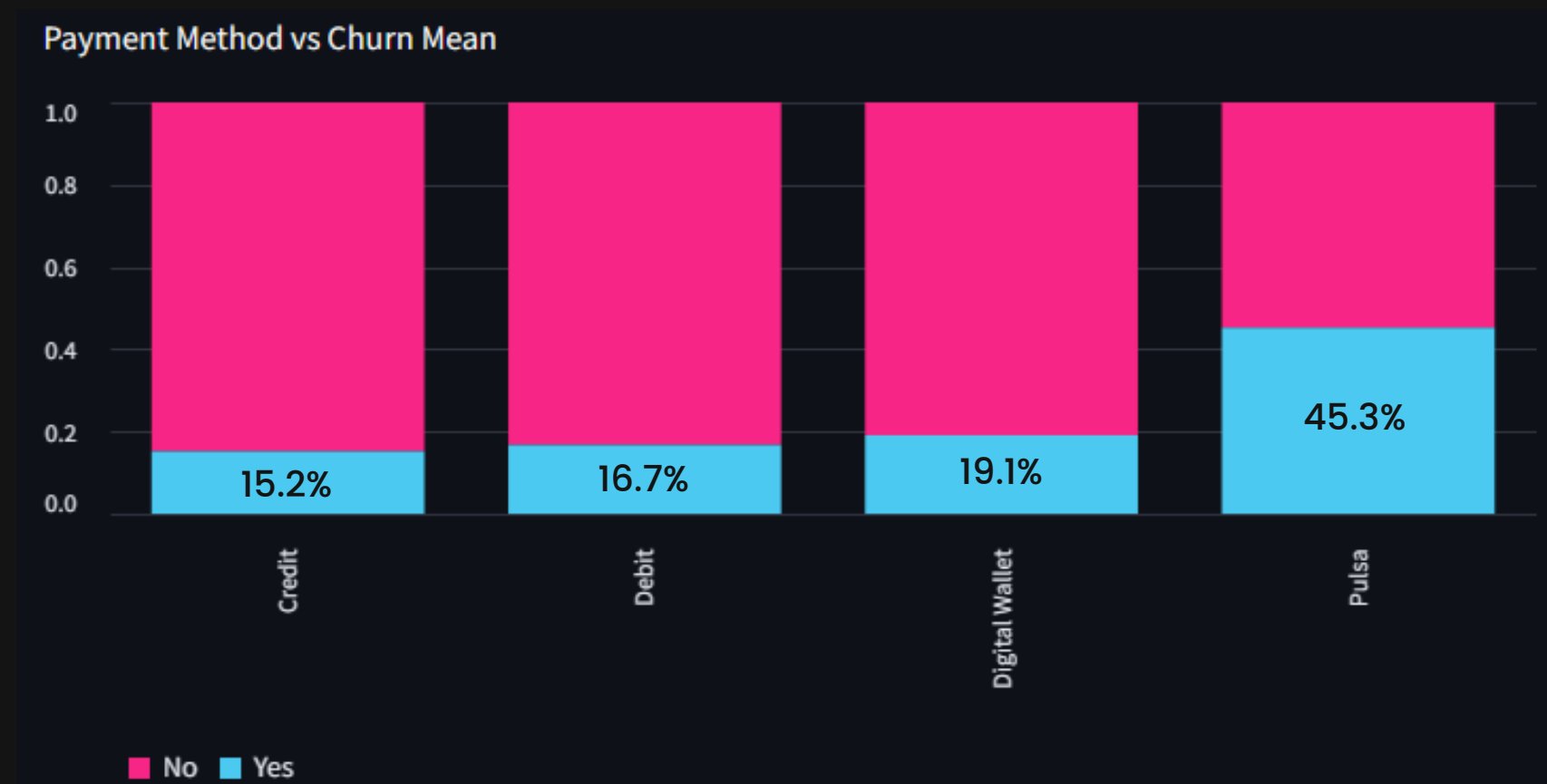




Correlation Analysis

Some has Moderate Correlation to Churn!

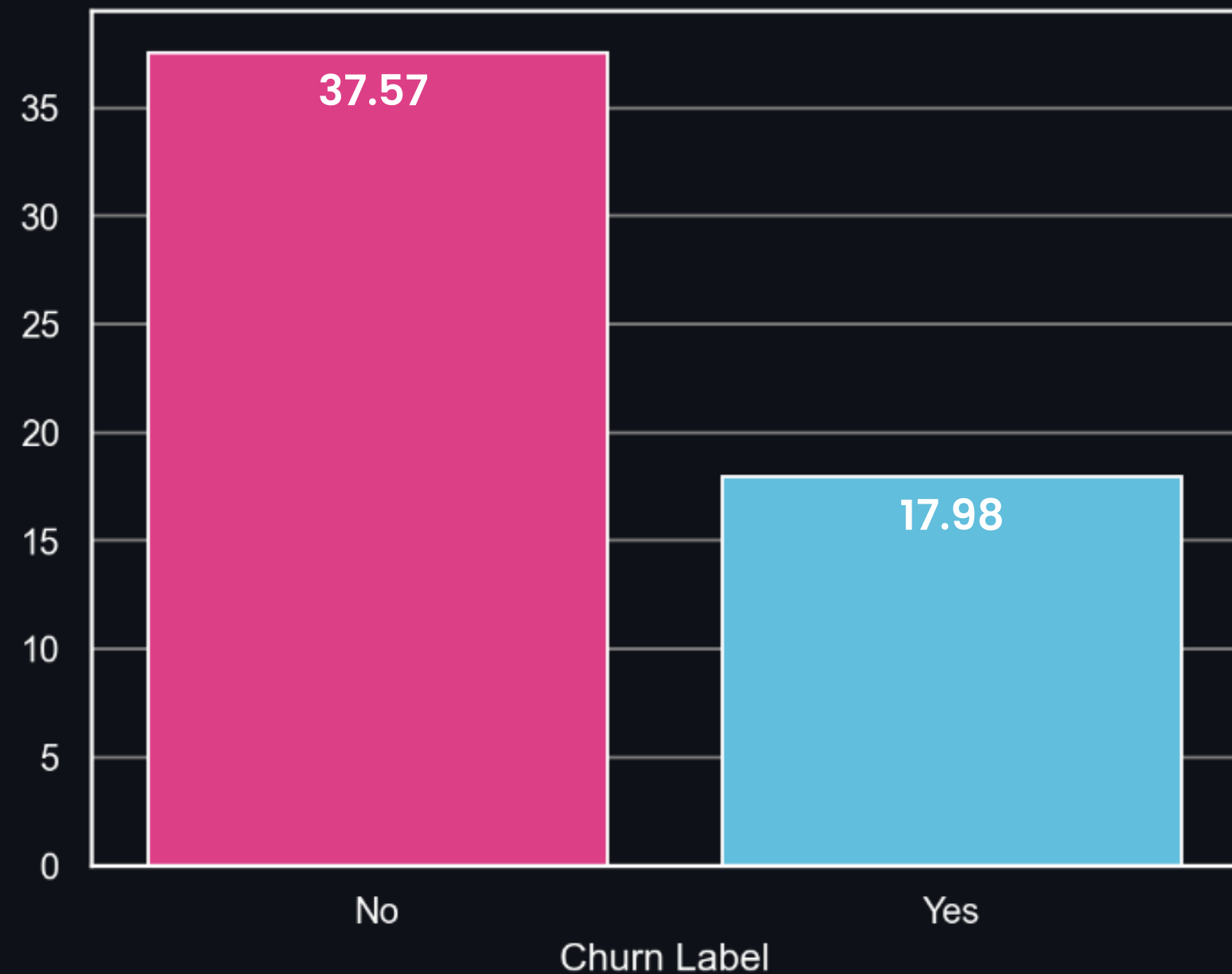
- Device Class (0.26)
- Tenure Months (0.38)
- pm_pulsa (0.29)



Customer Loyalty Trend Analysis

We can observe from the graph that customers who use high-end devices and those who use "pulsa" are more likely to churn.

Tenure Months Mean vs Churn

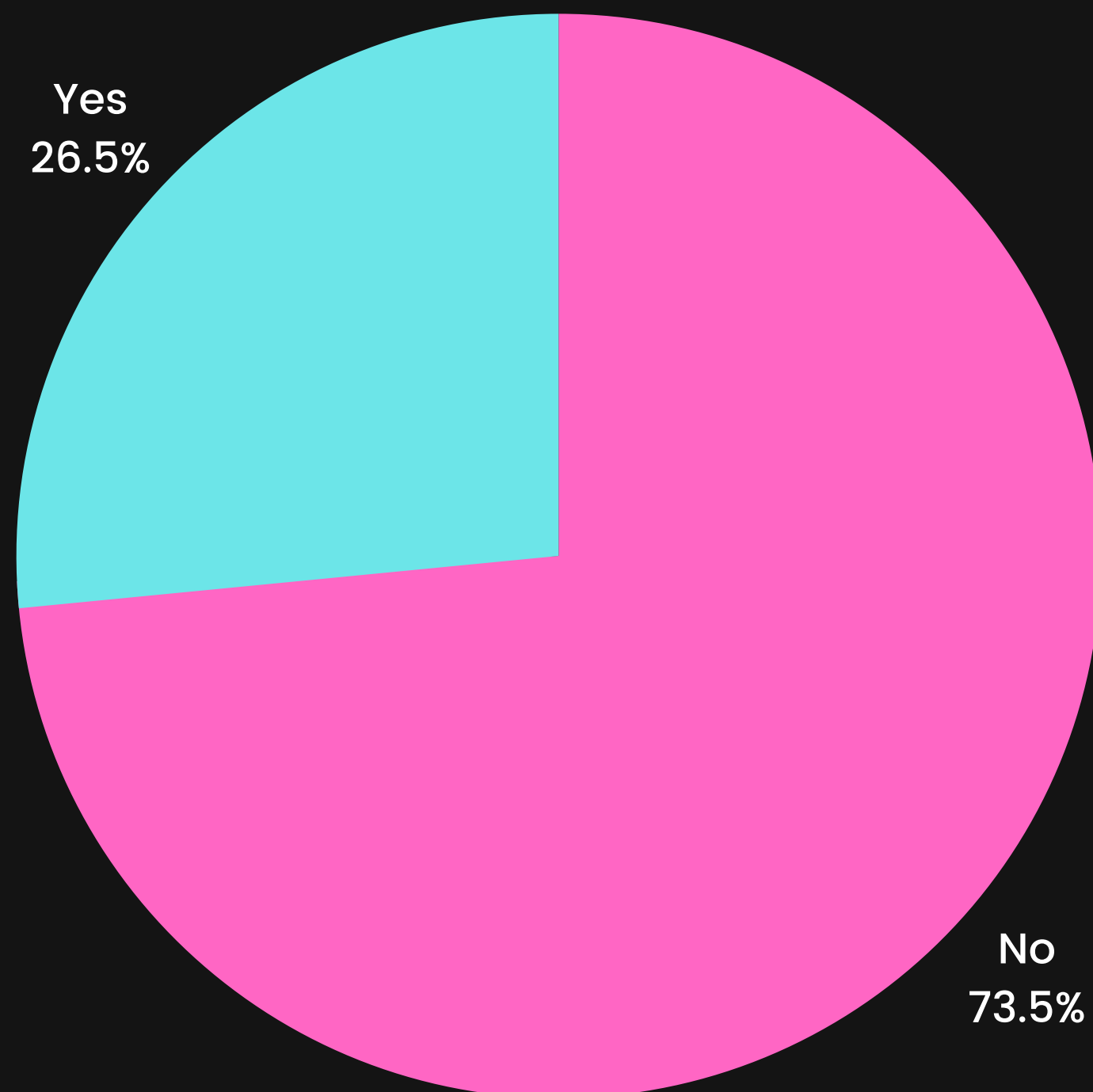


Customer Loyalty Trend Analysis

From the graph, it can be observed that the longer a customer has been with the company, the less likely they are to churn.



Churn Label Distribution

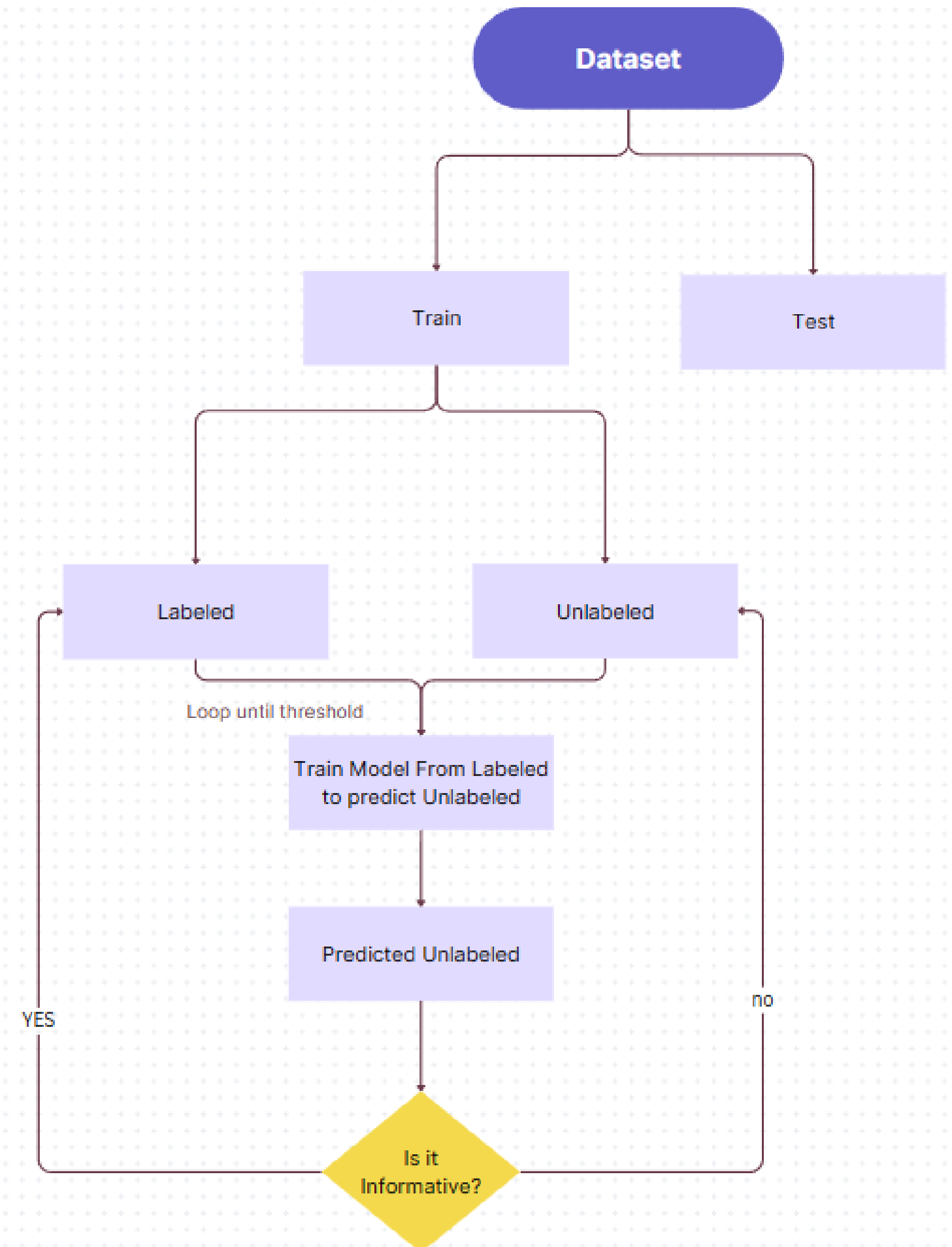
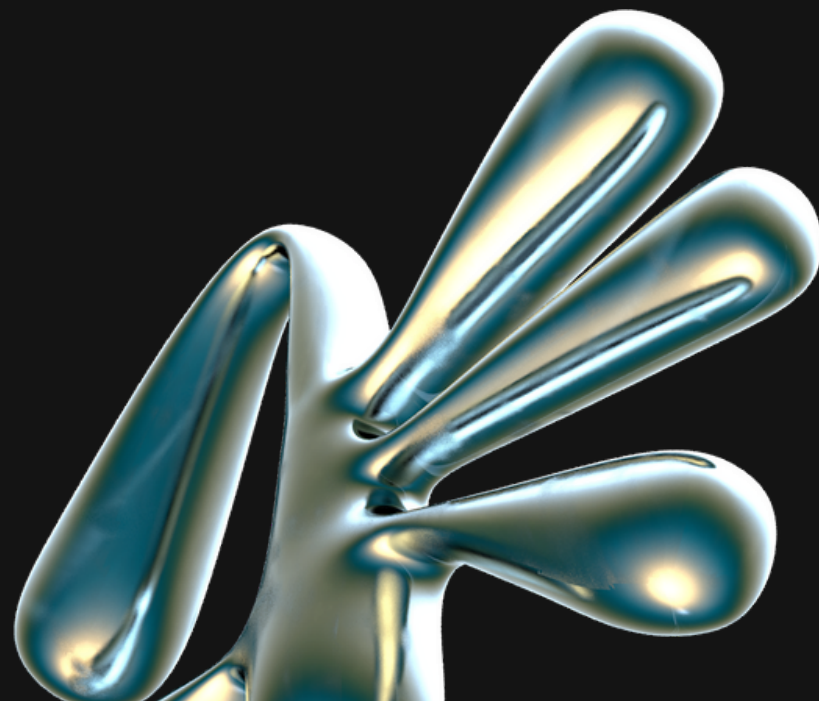


Problem?

It's highly unbalanced!

Active Learning Flow

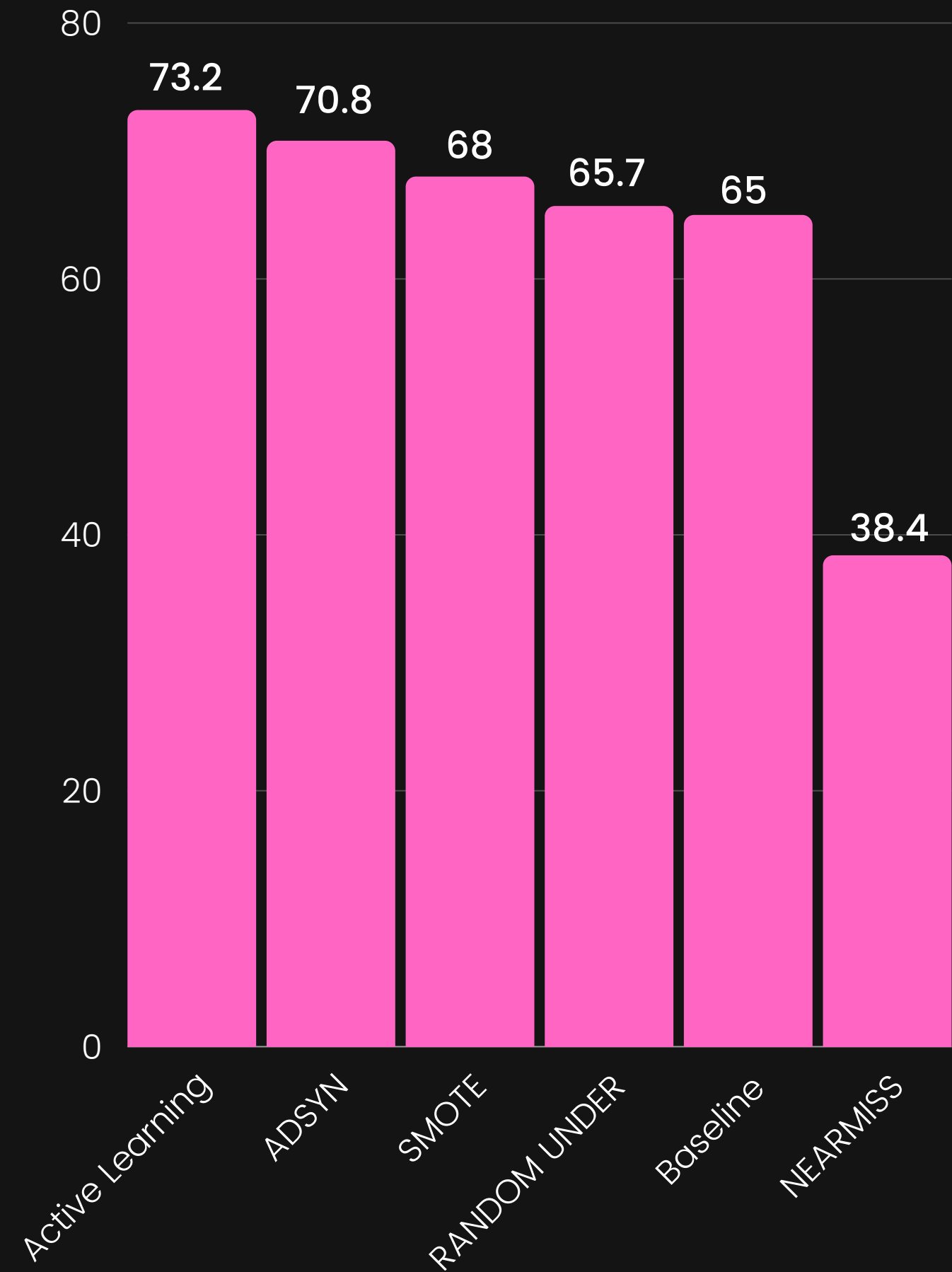
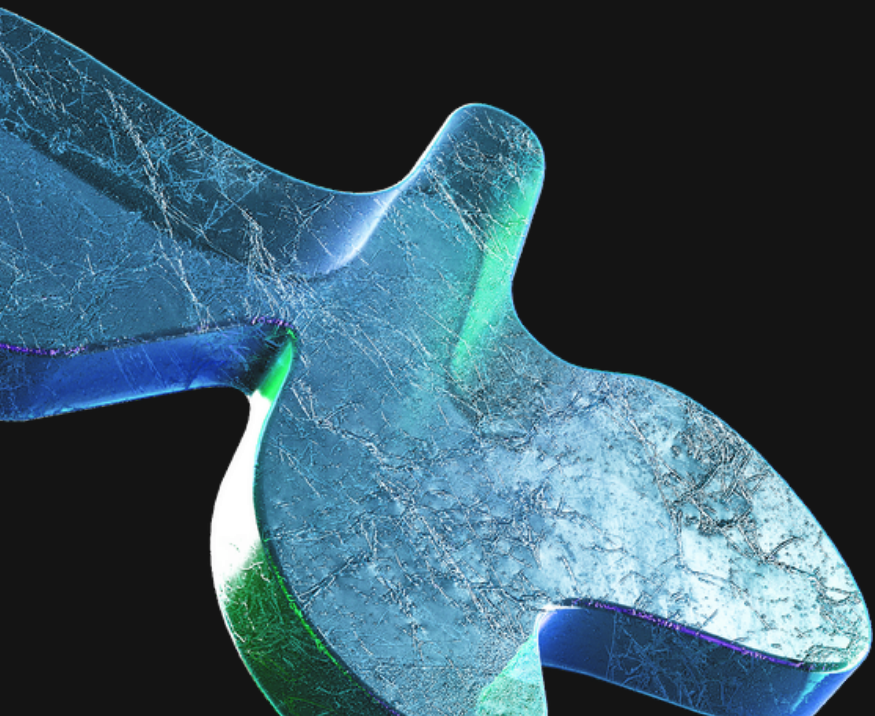
*The labeled data at the start is filled with the minority class.



F1 differences

According to the dataset, active learning appears to be the most effective approach, exhibiting a notable 12% increase in F1 score when compared to alternative methods.

Model used: CatBoost



Deploy

CLTV (Predicted Thou. IDR)

3227.90

Predict

Information

How does it work ?

Complete all the parameters and the machine learning model will predict the probability that the customer will churn and generate the possible reason and solution.

Results

Churn Probability

The probability that the customer will churn is 98.64%

Reason and Solution

Reason: Based on the given data, the possible reason that the customer left the company in this quarter could be the lack of usage of internet services for games, music, education, and MyApp. The customer has also not utilized the call center service, indicating a lack of interaction or support from the company. Additionally, the customer's device class is classified as "High End", suggesting



LLM? gpt-3.5 turbo

We utilize the power of the OpenAI API to generate reasons and effective solutions for various problems.

Streamlit

We employ Streamlit as the user-friendly application to facilitate prediction and provide a visually appealing dashboard for users to interact with.

LLM

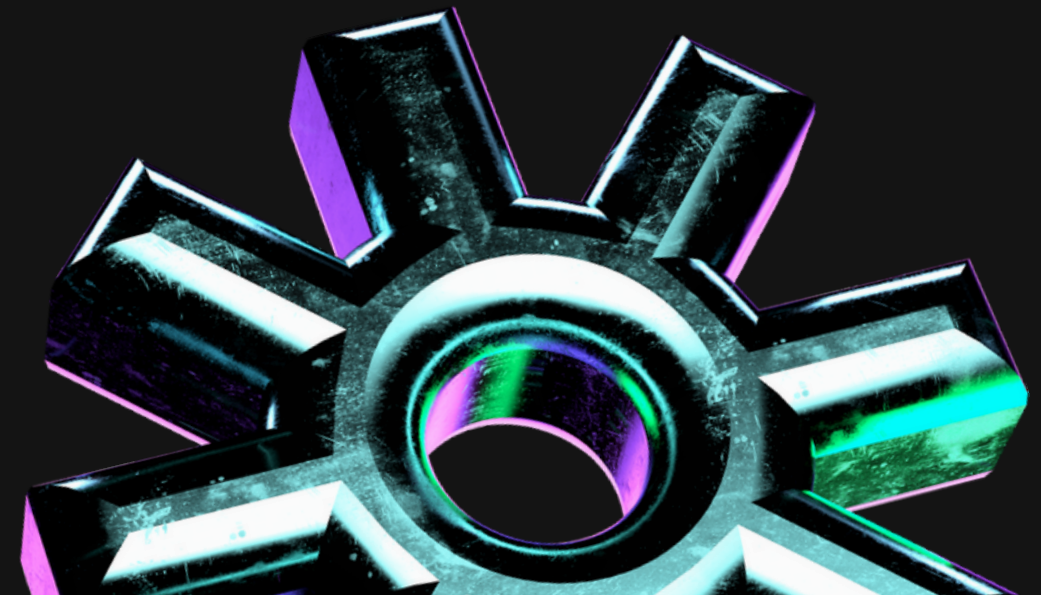
Prompt Engineering

Reason and Solution Prompt

- Say you are a solution analyst. A telecommunication company wants to analyze its customer's behavior. Given a dataset that shows the usage of telecommunication services in Indonesia.
- There are several columns provided in the dataset:
 - [Column Name]: [Description]
- These are the unique value for all categorical columns:
 - [Column Name]: [Unique Values]
- Here is the row of data (in csv format):
 - [A row of input data]
- From the given data, predict the possible reason that the customer left the company in this quarter and provide the solution to prevent the customer to leave the company!
- Make it in this format:
 - Reason: ...
 - Solution: ...

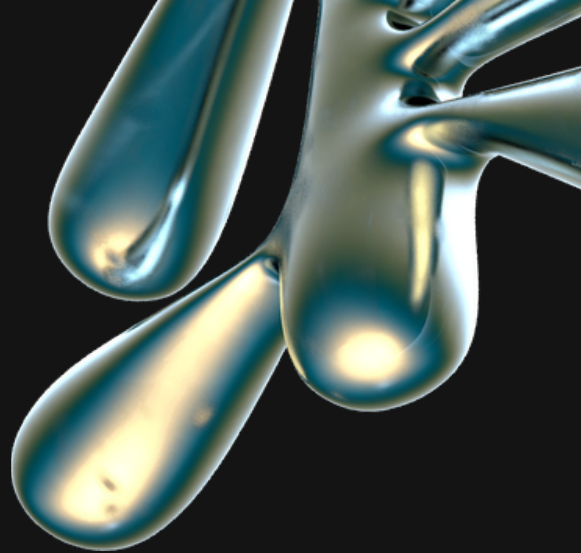
Analyze Data Prompt

- Analyze these data:
- # [Column Name] Mean vs Churn
- {data_mean_vs_churn.to_csv()}
- Give me the insights for telecommunication company from the data in this format in points and maximum 5 points:
 - Insights:
 - 1. ...
 - 2. ...
 - ...



Reason and Solution

LLM Generation Example



Input

Device Class	High End
Use Call Center?	No
Use Games Product?	No
Use Music Product?	No
Use Education Product?	No
Use Video Product?	Yes

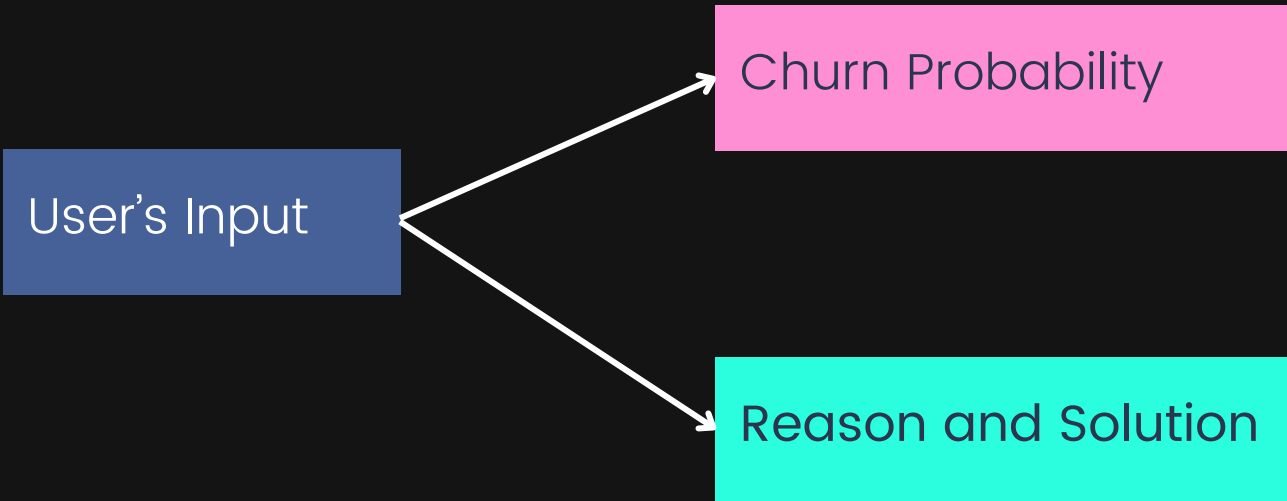
Device Class	High End
Use MyApp?	No
Payment Method	Pulsa
Tenure Months	1
Monthly Purchase (Thou. IDR)	103155
CLTV (Predicted Thou. IDR)	3227.90

Output

- Reason: Based on the given data, the possible reason that the customer left the company in this quarter could be the absence of internet service for games, music, and education products. The customer's device class is High End, indicating that the customer is likely to have higher expectations for internet services. However, the customer is not using any of the internet services for games, music, and education. This might have led to dissatisfaction and the decision to leave the company.
- Solution: To prevent customers from leaving the company, the telecom company should focus on providing a comprehensive range of internet services, including games, music, and education products, to cater to the diverse needs of its customers. By offering these additional services, the company can enhance customer satisfaction and provide a more compelling reason for customers to stay. Additionally, the company should communicate and educate customers about the availability and benefits of these services, ensuring that customers are aware of the full range of options. Offering attractive incentives or discounts for customers who utilize multiple services can also help incentivize customers to continue their subscription with the company.

Application Demo

Tab 1: Predict



Strategic Customer Retention

Discovering the Potency of Active Learning on Imbalanced Data, Fused with Personalized LLM-Driven Churn Solutions

Predict Analysis

Find out if your customer will churn or not

Device Class

High End

Use Call Center?

No

Has Internet Service?

Yes

Use Games Product?

No

Use Music Product?

No

Education Product

No

Use Video Product?

Yes

Use MyApp?

How does it work ?

Complete all the parameters and the machine learning model will predict the probability that the customer will churn and generate the possible reason and solution.

Information

Customers are a crucial aspect of any telecommunication business. This is a system that can offer solutions to prevent customers from leaving the company.

The model utilized is CatBoost, which has been trained using a active learning downsampling technique.

Tenure Months

1

Monthly Purchase (Thou. IDR)

103155

CLTV (Predicted Thou. IDR)

3227.90

Predict

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Churn Probability

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Reason and Solution

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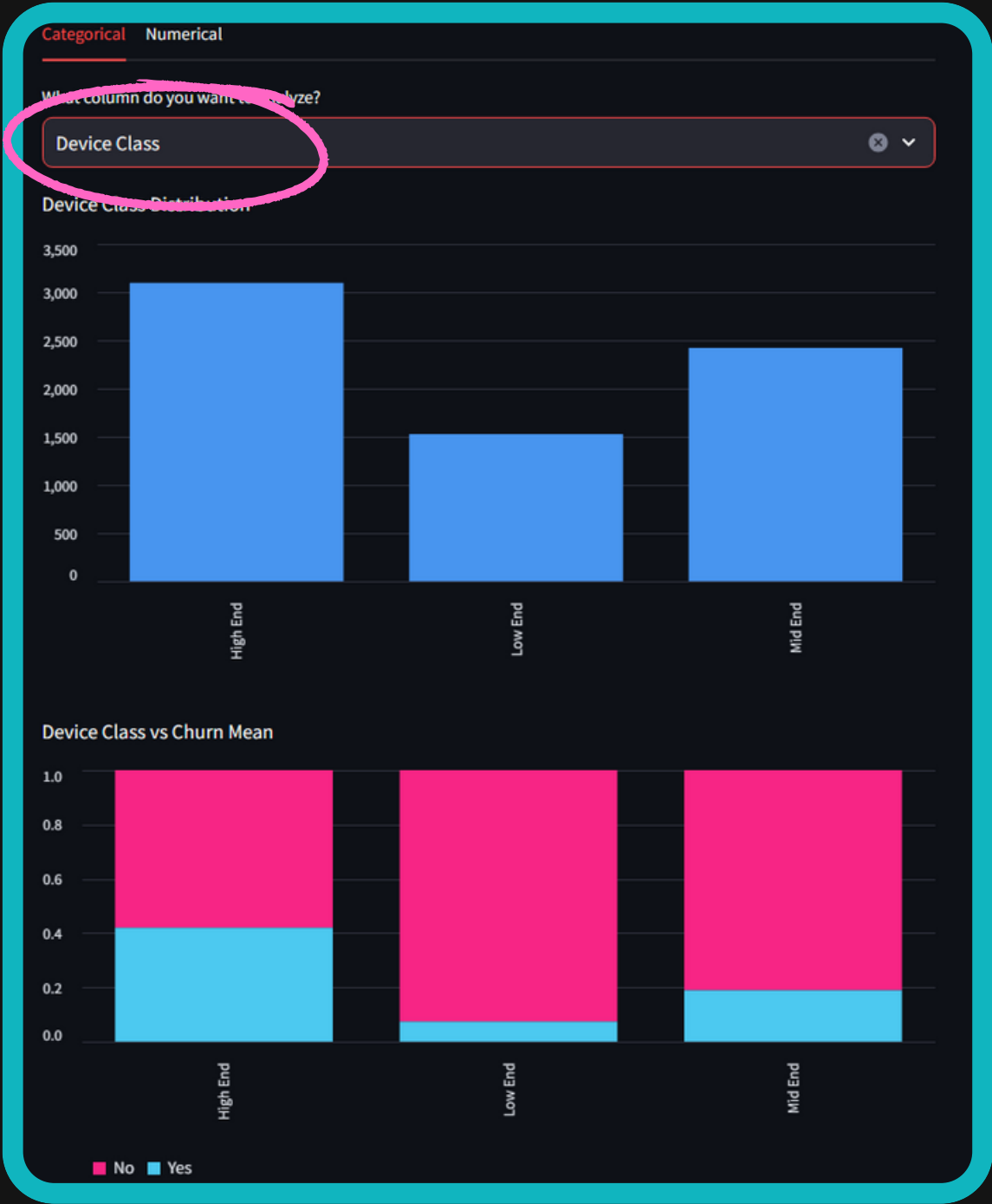
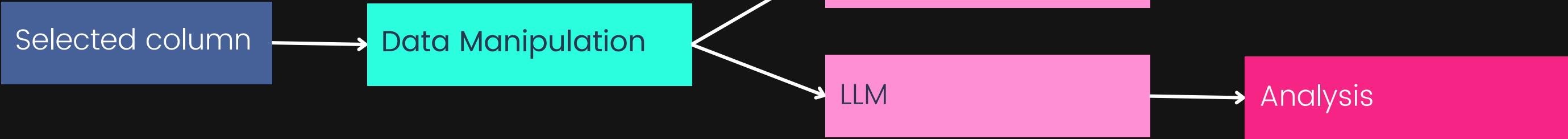
Complete the parameters!

Click Predict!

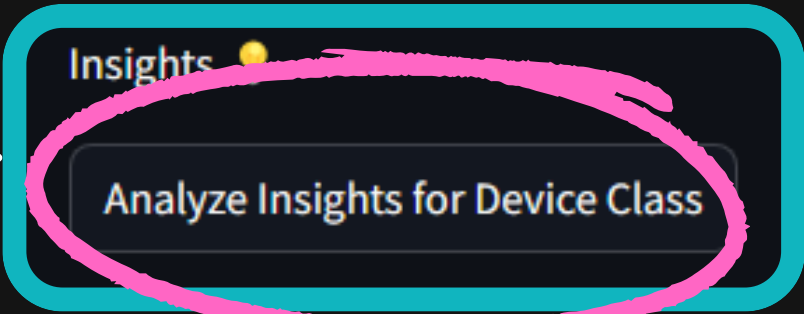
Predicted!

Application Demo

Tab 2: Analysis



Select a column



Click Analyze insights!

Insights

Analyze Insights for Device Class

Insights for telecommunication company from the given data:

1. High-end devices have the highest count (3096), followed by Mid-end devices (2421), and Low-end devices (1526). This suggests that the majority of customers prefer to use high-end devices.
2. The churn rate for High-end devices is almost evenly distributed, with 58.1% of customers not churning and 41.9% churning. This indicates that customers using high-end devices are slightly more likely to churn compared to other device classes.
3. Low-end devices have the highest churn rate, with only 7.4% of customers churning. This implies that customers using low-end devices are the least likely to churn.
4. Mid-end devices show a relatively balanced churn rate, with 81.0% of customers not churning and 18.9% churning. This suggests that customers using mid-end devices have an average likelihood of churning.
5. Overall, there is a correlation between the device class and churn rate, indicating that device class may be a significant factor influencing customer churn. Telecommunication company should consider offering incentives or special offers to customers using low or mid-end devices to enhance customer retention.

Analyzed!



Try it yourself!

<https://threeoutliers-dsw.streamlit.app/>

Conclusion

In summary, our customer churn prediction model leverages active learning and outperforms conventional techniques such as SMOTE and ADASYN in handling imbalanced data. Notably, it achieves a higher F1 score based on the given dataset.

The integration of Streamlit enhances user interaction with a visually appealing dashboard. Incorporating LLM for solution generation based on customer information offers a comprehensive approach. Our solution stands as an advanced and practical application of machine learning, empowering businesses to proactively address churn challenges and make informed decisions for effective customer retention.



Future Works

Enhance data collection by incorporating user reasons, then leverage NLP to extract features. This can be utilized to fine-tune the Language Model (LLM) for a specific task, enhancing the quality of the generated solutions based on the extracted features.

Develop a feature allowing the model to accept input in CSV format, generating reasons and solutions for rows with churn probabilities exceeding 0.5. However, it's important to note that this would require substantial resources for the LLM due to the increased computational demands associated with processing a larger volume of data.

Thank you!

